



Atezolizumab plus nab-paclitaxel as first-line treatment for unresectable, locally advanced or metastatic triple-negative breast cancer (IMpassion130): updated efficacy results from a randomised, double-blind, placebo-controlled, phase 3 trial

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Summary

Background Immunotherapy in combination with chemotherapy has shown promising efficacy across many different tumour types. We report the prespecified second interim overall survival analysis of the phase 3 IMpassion130 study assessing the efficacy and safety of atezolizumab plus nab-paclitaxel in patients with unresectable, locally advanced or metastatic triple-negative breast cancer.

Methods In this randomised, placebo-controlled, double-blind, phase 3 trial, done in 246 academic centres and community oncology practices in 41 countries, patients aged 18 years or older, with previously untreated, histologically documented, locally advanced or metastatic triple-negative breast cancer, and Eastern Cooperative Oncology Group performance status of 0 or 1 were eligible. Patients were randomly assigned (1:1) using a permuted block method (block size of four) and an interactive voice–web response system. Randomisation was stratified by previous taxane use, liver metastases, and PD-L1 expression on tumour-infiltrating immune cells. Patients received atezolizumab 840 mg or matching placebo intravenously on day 1 and day 15 of every 28-day cycle and nab-paclitaxel 100 mg/m² of body surface area intravenously on days 1, 8, and 15 until progression or unacceptable toxicity. Investigators, patients, and the funder were masked to treatment assignment. Coprimary endpoints were investigator-assessed progression-free survival per Response Evaluation Criteria in Solid Tumors version 1.1 and overall survival, assessed in the intention-to-treat population and in patients with PD-L1 immune cell-positive tumours (tumours with ≥1% PD-L1 expression). The final progression-free survival results were previously reported at the first interim overall survival analysis. The prespecified statistical testing hierarchy meant that overall survival in the subgroup of PD-L1 immune cell-positive patients could only be formally tested if overall survival was significantly different between the treatment groups in the intention-to-treat population. This study is registered with ClinicalTrials.gov, NCT02425891.

Findings Between June 23, 2015, and May 24, 2017, 902 patients were enrolled, of whom 451 were randomly assigned to receive atezolizumab plus nab-paclitaxel and 451 were assigned to receive placebo plus nab-paclitaxel (the intention-to-treat population). Six patients from each group did not receive treatment. At the second interim analysis (data cutoff Jan 2, 2019), median follow-up was 18·5 months (IQR 9·6–22·8) in the atezolizumab group and 17·5 months (8·4–22·4) in the placebo group. Median overall survival in the intention-to-treat patients was 21·0 months (95% CI 19·0–22·6) with atezolizumab and 18·7 months (16·9–20·3) with placebo (stratified hazard ratio [HR] 0·86, 95% CI 0·72–1·02, *p*=0·078). In the exploratory overall survival analysis in patients with PD-L1 immune cell-positive tumours, median overall survival was 25·0 months (95% CI 19·6–30·7) with atezolizumab versus 18·0 months (13·6–20·1) with placebo (stratified HR 0·71, 0·54–0·94). As of Sept 3, 2018 (the date up to which updated safety data were available), the most common grade 3–4 adverse events were neutropenia (38 [8%] of 453 patients in the atezolizumab group vs 36 [8%] of 437 patients in the placebo group), peripheral neuropathy (25 [6%] vs 12 [3%]), decreased neutrophil count (22 [5%] vs 16 [4%]), and fatigue (17 [4%] vs 15 [3%]). Treatment-related deaths occurred in two (<1%) patients in the atezolizumab group (autoimmune hepatitis related to atezolizumab [*n*=1] and septic shock related to nab-paclitaxel [*n*=1]) and one (<1%) patient in the placebo group (hepatic failure). No new treatment-related deaths have been reported since the primary clinical data cutoff date (April 17, 2018).

Interpretation Consistent with the first interim analysis, this second interim overall survival analysis of IMpassion130 indicates no significant difference in overall survival between the treatment groups in the intention-to-treat population but suggests a clinically meaningful overall survival benefit with atezolizumab plus nab-paclitaxel in patients with PD-L1 immune cell-positive disease. However, this positive result could not be formally tested due to the prespecified statistical testing hierarchy. For patients with PD-L1 immune cell-positive metastatic triple-negative breast cancer, atezolizumab plus nab-paclitaxel is an important therapeutic option in a disease with high unmet need.

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Introduction

Triple-negative breast cancer constitutes 15–20% of cases of breast cancer and is defined by the absence of oestrogen receptors, progesterone receptors, and over-expression or gene amplification of HER2 on the surface of cancer cells.¹ Until the first report of the IMpassion130 trial,² chemotherapy was the standard of care for first-line systemic treatment for patients with triple-negative breast cancer that has advanced or metastasised.^{3–5} In several countries, such as Japan, and in Europe, other approved treatment options are bevacizumab in combination with chemotherapy and poly (ADP-ribose) polymerase inhibitors (eg, olaparib and talazoparib) for patients with *BRCA*-mutant, HER2-negative tumours.^{6–8} Prognoses remain poor, with reported median overall

survival estimates of approximately 18 months or less with available treatments. Because of the promising efficacy of checkpoint inhibition in other tumour types,^{9–13} inhibitors of PD-L1 and PD-1 as monotherapy or in combination with chemotherapy have been investigated in triple-negative breast cancer.^{14–16} The higher prevalence of PD-L1 expression in triple-negative breast cancer compared with hormone receptor-positive breast cancer subtypes further supports this therapeutic approach.^{17–21}

The phase 3 study, IMpassion130,² evaluated atezolizumab, a monoclonal antibody targeting PD-L1, plus nab-paclitaxel compared with placebo plus nab-paclitaxel as a first-line treatment for patients with unresectable locally advanced or metastatic triple-negative breast cancer. At the time of primary analysis of IMpassion130, the final

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Research in context

Evidence before this study

We searched PubMed for clinical trials published between June 30, 2009, and July 30, 2019, in any language using the search terms “PD-L1” or “PD-1” or “immunotherapy” or “immune checkpoint” in combination with “TNBC” or “triple-negative breast cancer”. We identified a feasibility study of a personalised peptide vaccine and a phase 1–2 study planning to enrol patients with triple-negative breast cancer to evaluate an RNA vaccine. Several studies reported increased expression of PD-L1 and PD-1 in triple-negative breast cancer, suggesting that inhibition of the PD-L1–PD-1 immune checkpoint could result in clinical benefit in this disease. Promising responses and duration of response in phase 1 studies evaluating single-drug atezolizumab, pembrolizumab, or avelumab had been reported. A tolerable safety profile and increased proportions of patients achieving a response compared with single-drug approaches had been reported for the combination of atezolizumab plus nab-paclitaxel in a phase 1b study.

Added value of this study

To our knowledge, IMpassion130 is the first reported phase 3 study that evaluated an antibody targeting the PD-L1–PD-1 immune checkpoint in patients with unresectable locally advanced or metastatic triple-negative breast cancer. The study met its coprimary progression-free survival endpoint in the primary analysis, showing a significant progression-free survival benefit with atezolizumab plus nab-paclitaxel compared with placebo plus nab-paclitaxel as a first-line treatment for patients with advanced or metastatic triple-negative breast cancer. Because no treatment effect was seen in the PD-L1 immune cell-negative subgroup, the clinical benefit in the intention-to-treat population was driven by the positive results in the PD-L1 immune cell-positive subgroup of patients. The first interim overall survival analysis (concurrent with the primary progression-free survival analysis) showed no significant difference in overall survival between the treatment groups in the intention-to-treat population but a clinically meaningful

improvement in median overall survival in patients with PD-L1 immune cell-positive disease. This preplanned second interim overall survival analysis, which was done after 534 (59%) of 902 deaths had occurred in the intention-to-treat population (80% information fraction) also showed no significant difference in overall survival between the groups in the intention-to-treat population. Although the overall survival results provide evidence for the clinical benefit in patients with PD-L1 immune cell-positive disease, consistent with the primary analysis, this result could not be formally tested due to the prespecified statistical testing hierarchy and thus these findings can only be regarded as exploratory. The reported safety update supports an unchanged safety profile with atezolizumab plus nab-paclitaxel relative to the first interim analysis.

Implications of all the available evidence

The randomised controlled phase 3 IMpassion130 study did not suggest a significant overall survival benefit from the addition of atezolizumab to nab-paclitaxel in the intention-to-treat population of patients with unresectable, locally advanced or metastatic triple-negative breast cancer. However, providing evidence to support promising clinical activity observed in phase 1 studies evaluating drugs targeting PD-L1–PD-1 signalling, the study results did indicate clinically meaningful improvement in median progression-free and overall survival of atezolizumab plus nab-paclitaxel in patients with PD-L1 immune cell-positive unresectable locally advanced or metastatic triple-negative breast cancer. The findings of this second interim overall survival analysis, reported here, provide evidence that the clinical benefit is consistent and maintained with extended follow-up. Approvals by US and European Union regulatory agencies, and the inclusion of this regimen in US National Comprehensive Cancer Network treatment guidelines, highlight that atezolizumab plus nab-paclitaxel could be a new standard of care for previously untreated patients with PD-L1 immune cell-positive, unresectable, locally advanced or metastatic triple-negative breast cancer.

progression-free survival analysis and the first interim overall survival analysis were done (clinical data cutoff April 17, 2018). The study met its coprimary progression-free survival endpoints, showing clinical benefit with atezolizumab plus nab-paclitaxel when compared with placebo plus nab-paclitaxel in both the intention-to-treat population and the subgroup of patients with tumours that express PD-L1 on immune cells that cover 1% or more of the tumour area (PD-L1 immune cell-positive patients).² The treatment effect in the PD-L1 immune cell-positive patient subgroup was a clinically meaningful benefit: the median progression-free survival was 7.5 months (95% CI 6.7–9.2) with atezolizumab plus nab-paclitaxel versus 5.0 months (3.8–5.6) with placebo plus nab-paclitaxel (hazard ratio [HR] 0.62, 95% CI 0.49–0.78], $p < 0.0001$).² Based on these final progression-free survival results, the US Food and Drug Administration granted accelerated approval on March 8, 2019, to atezolizumab plus nab-paclitaxel as a first-line treatment for patients with unresectable locally advanced or metastatic triple-negative breast cancer with tumours that express PD-L1 on immune cells that cover 1% or more of the tumour area (PD-L1 immune-cell positive).²² The European Commission has also approved atezolizumab plus nab-paclitaxel for the treatment of adult patients with unresectable locally advanced or metastatic triple-negative breast cancer with tumours that are PD-L1 positive ($\geq 1\%$ PD-L1 expression) and who have not received previous chemotherapy for metastatic disease.²³

As previously reported,² the first interim overall survival analysis and concurrent final progression-free survival analysis were done after 43% of deaths in the intention-to-treat population had occurred, with a median follow-up of 12.9 months (IQR 8.8–16.5) and 59% information fraction, which is the percentage of the total information expected at the scheduled end of the trial. With an HR for overall survival of 0.84 (95% CI 0.69–1.02, $p = 0.08$) in the intention-to-treat population, the first interim overall survival analysis did not cross the prespecified significance boundary ($\alpha = 0.0065$). A Kaplan-Meier analysis of overall survival in the PD-L1 immune cell-positive subgroup that was not formally tested because of the prespecified statistical testing hierarchy (since significance in the intention-to-treat population was required to enable subsequent testing in the PD-L1 immune cell-positive subgroup) showed a median overall survival of 25.0 months (95% CI 22.6–not estimable) with atezolizumab plus nab-paclitaxel versus 15.5 months (13.1–19.4) with placebo plus nab-paclitaxel (HR 0.62, 95% CI 0.45–0.86).²

In this paper, we report the prespecified second interim overall survival analysis from the IMpassion130 study.

Methods

Study design and participants

The IMpassion130 trial is a randomised, double-blind, placebo-controlled trial that was done in 246 academic

centres and community oncology practices in 41 countries in Europe, North America, Asia, and Latin America (appendix pp 2–6). Full details of the study design and methods have previously been published.²¹ Patients were aged 18 years or older and had histologically documented, unresectable, locally advanced or metastatic triple-negative breast cancer and a life expectancy of at least 12 weeks. HER2-negative, oestrogen receptor-negative, and progesterone receptor-negative status was evaluated by local institutions before enrolment per guidelines of the American Society of Clinical Oncology and the College of American Pathologists.^{24,25} Other key inclusion criteria were that patients had to provide representative tumour specimens that were evaluable for prospective testing of PD-L1 expression by immunohistochemistry (SP142 PD-L1 immunohistochemical assay [Ventana Medical Systems, Oro Valley, AZ, USA]; they were eligible for taxane monotherapy; they had no previous chemotherapy or targeted therapy for metastatic triple-negative breast cancer; they had measurable disease per Response Evaluation Criteria in Solid Tumors (RECIST) version 1.1; they had an Eastern Cooperative Oncology Group (ECOG) performance status score of 1 or less; and they had adequate organ and haematological function. Radiotherapy and previous curative chemotherapy completed 12 months or more before randomisation was allowed. Patients with treated asymptomatic CNS metastases were also allowed. Key exclusion criteria included untreated CNS disease; previous history of autoimmune disease; recent treatment (ie, within 4 weeks or five half-lives of the drug [whichever was shorter] before randomisation) with a systemic immunostimulatory drug; use of glucocorticoids or immunosuppressive drugs; and previous immune checkpoint-targeting therapies.

The trial was done in accordance with Good Clinical Practice guidelines and the Declaration of Helsinki. Protocol approval was obtained from independent review boards or ethics committees for each site. An independent data monitoring committee reviewed unmasked safety and study conduct data every 6 months. All patients provided written informed consent.

Our study protocol has been published online and is also available in the appendix.

Randomisation and masking

The principal investigators identified and enrolled patients, as described previously.² Using a permuted block method (with a block size of four) and an interactive voice-web response system, we randomly assigned patients in a 1:1 ratio to receive atezolizumab plus nab-paclitaxel or placebo plus nab-paclitaxel. We stratified randomisation by the presence of liver metastases (yes vs no), previous neoadjuvant or adjuvant taxane therapy (yes vs no), and PD-L1 expression on tumour-infiltrating immune cell as a percentage of tumour area ($< 1\%$ [PD-L1 immune cell negative] vs $\geq 1\%$ [PD-L1 immune cell

For the protocol see
https://www.nejm.org/doi/suppl/10.1056/NEJMoa1809615/suppl_file/nejm1809615_protocol.pdf

positive)). The investigators, study site personnel, and patients were masked to treatment assignment. Masking was achieved by ensuring that the placebo was not distinguishable from atezolizumab.

Procedures

Patients received either atezolizumab 840 mg or matching placebo on days 1 and 15 of every 28-day treatment cycle and nab-paclitaxel 100 mg/m² of body surface area on days 1, 8, and 15 of every cycle. All drugs were administered intravenously, and patients received study treatments until disease progression, as assessed by the investigators per RECIST 1.1, or occurrence of unacceptable toxicity (according to the medical assessment of the treating investigator who was directly caring for the patient). If no unacceptable toxicity occurred, nab-paclitaxel was administered for at least six cycles of treatment. Dose reduction of atezolizumab or placebo was not permitted, but treatment could be temporarily suspended for as long as deemed necessary by the treating investigator, if required to manage an adverse event (of any severity); decisions regarding suspension of treatment were made on an individual basis by the treating investigator and medical monitor by assessing the benefits and risks to that patient. The treating physician could use discretion in modifying or accelerating the dose modification guidelines depending on the severity of toxicity and an assessment of the risk versus benefit for the patient, with the goal of maximizing patient compliance and access to supportive care. The dose of nab-paclitaxel could be reduced to 75 mg/m², 50 mg/m², and then discontinued, if required because of haematological toxicity (of any severity). The treating investigator assessed patients for adverse events (according to the Common Terminology Criteria for Adverse Events [version 4.0] of the National Cancer Institute) and laboratory values (serum chemistry) on days 1, 8, and 15 of every 28-day cycle. Atezolizumab or nab-paclitaxel could be discontinued independently in the absence of disease progression based on RECIST 1.1. The investigators imaged tumours by CT or MRI at baseline, every 8 weeks for 12 months, then every 12 weeks until disease progression. We followed up patients for survival every 3 months after treatment discontinuation until death.

Outcomes

The coprimary efficacy endpoints were progression-free survival, assessed by investigators per RECIST 1.1, and overall survival. These endpoints were evaluated in the intention-to-treat population and the PD-L1 immune cell-positive subgroup of patients. Prespecified secondary efficacy endpoints were the proportion of patients achieving an objective response (ie, a complete or partial response according to RECIST 1.1) and duration of response (measured from the first objective response to disease progression per RECIST version 1.1 or death from

any cause, whichever occurred first), and time to deterioration in global health status (in items 29 and 30 of the European Organisation for Research and Treatment of Cancer Quality of life Questionnaire Core 30). Quality-of-life outcomes will be reported in a separate publication. Exploratory endpoints included predictive, prognostic, and pharmacodynamic biomarkers and their association with disease status and response to study treatment. Exploratory analyses of the distribution of PD-L1-positive immune cells and its interaction as a continuous variable and the activity of atezolizumab plus nab-paclitaxel in patients with PD-L1 expression in tumour cells are part of a separate biomarker analysis, which will be reported in a separate publication. Remaining exploratory endpoints were related to additional patient-reported outcomes, including health economic data, which will be reported elsewhere. Definitions for prespecified adverse events of special interest, which might be suggestive of an immune-mediated disorder, are summarised in the protocol.

Statistical analysis

The trial was originally designed to enrol 350 patients, with a primary progression-free survival endpoint. On Nov 20, 2015, the study protocol was amended to increase the sample size to approximately 900 patients (intention-to-treat population), supporting the promotion of overall survival from a secondary endpoint to a coprimary endpoint (protocol version 4). Progression-free survival and overall survival were analysed by intention-to-treat and in the PD-L1 immune cell-positive population. Two interim analyses and a final overall survival analysis were planned. The first interim analysis for overall survival was done with an information fraction of 59% (ie, when 59% of the total information expected at the scheduled end of the trial was available, after approximately 352 deaths had occurred); the definitive progression-free survival analysis was done concurrently.²¹ The second interim analysis for overall survival was planned with an information fraction of 80% available (available alpha was 0.0212 obtained from an alpha spending function). Overall survival and progression-free survival were compared between treatment groups using a stratified log-rank test, and HRs for death and disease progression were estimated using a stratified Cox proportional hazards model. The analysis was stratified by the same stratification factors that were used for the randomisation. The type I error for the study ($\alpha=0.05$) was controlled for treatment comparisons of the coprimary endpoints of progression-free survival and overall survival and for the secondary efficacy endpoint of objective responses in the intention-to-treat and PD-L1 immune cell-positive populations, using a stepwise testing procedure. Post-hoc exploratory results in the PD-L1 immune cell-negative population are displayed for comparison. The interim analysis significance boundary was based on the Lan-DeMets implementation of the O'Brien-Fleming use function. The null hypothesis of no difference in overall

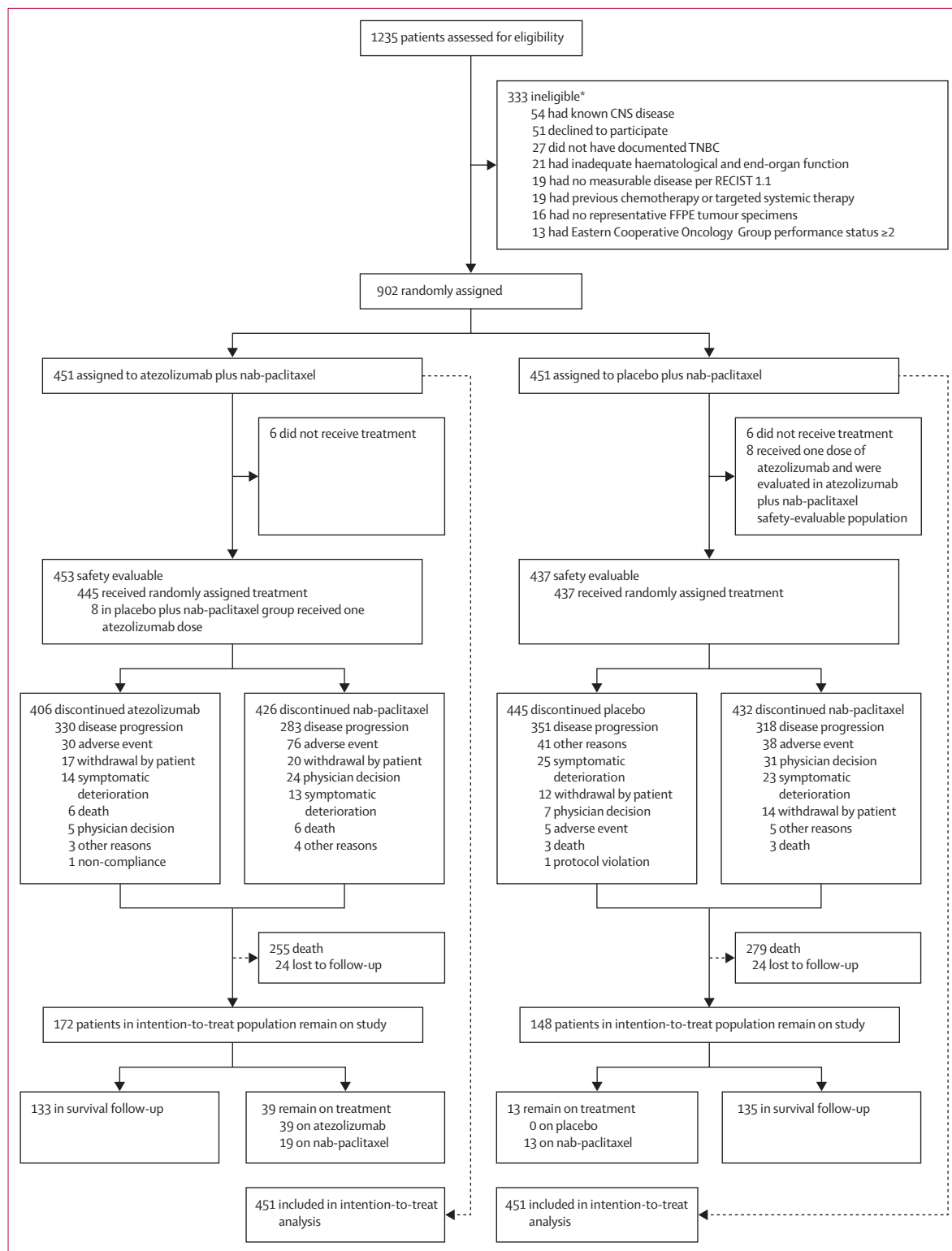


Figure 1: Trial profile

FFPE=formalin fixed and paraffin embedded. RECIST 1.1=Response Evaluation Criteria in Solid Tumors version 1.1. TNBC=triple-negative breast cancer. *Most common reasons for ineligibility.

	Intention-to-treat population (n=902)		PD-L1 immune cell-positive population (n=369)	
	Atezolizumab plus nab-paclitaxel group (n=451)	Placebo plus nab-paclitaxel group (n=451)	Atezolizumab plus nab-paclitaxel group (n=185)	Placebo plus nab-paclitaxel group (n=184)
Age, years	55 (46–64)	56 (47–65)	53 (44–63)	53 (44–63)
Age group, years				
18–40	63 (14%)	51 (11%)	31 (17%)	24 (13%)
41–64	284 (63%)	285 (63%)	111 (60%)	117 (64%)
≥65	104 (23%)	115 (25%)	43 (23%)	43 (23%)
Sex				
Female	448 (99%)	450 (>99%)	184 (99%)	184 (100%)
Male	3 (1%)	1 (<1%)	1 (1%)	0
Race				
White	308 (68%)	301 (67%)	125 (68%)	129 (70%)
Asian	85 (19%)	76 (17%)	38 (21%)	28 (15%)
Black or African-American	26 (6%)	33 (7%)	9 (5%)	14 (8%)
American Indian or Alaska Native	17 (4%)	23 (5%)	8 (4%)	9 (5%)
Native Hawaiian or other Pacific islander	1 (<1%)	0	0	0
Multiple	2 (<1%)	3 (1%)	0	0
Unknown	12 (3%)	15 (3%)	5 (3%)	4 (2%)
Eastern Cooperative Oncology Group performance status				
0	256/450 (57%)	270/450 (60%)	107/185 (58%)	112/184 (61%)
1	193/450 (43%)	179/450 (40%)	77/185 (42%)	72/184 (39%)
2*	1/450 (<1%)	1/450 (<1%)	1/185 (1%)	0
Metastatic disease	404/450 (90%)	408/450 (91%)	162/185 (88%)	159/183 (87%)
Number of metastatic sites				
0–3	332/450 (74%)	341/449 (76%)	149/185 (81%)	140/183 (77%)
≥4	118/450 (26%)	108/449 (24%)	36/185 (19%)	43/183 (23%)
Site of metastatic disease				
Liver†	126 (28%)	118 (26%)	44 (24%)	39 (21%)
Bone	145 (32%)	141 (31%)	54 (29%)	49 (27%)
Brain	30 (7%)	31 (7%)	15 (8%)	11 (6%)
Lung	226 (51%)	242 (54%)	86 (46%)	98 (53%)
Lymph node only	33/450 (7%)	23/449 (5%)	18/185 (10%)	13/183 (7%)
Previous neoadjuvant or adjuvant treatment	284 (63%)	286 (63%)	125 (68%)	117 (64%)
Previous taxane use†	231 (51%)	230 (51%)	96 (52%)	94 (51%)
Previous anthracycline use	243 (54%)	242 (54%)	109 (59%)	101 (55%)
Previous radiotherapy	268 (59%)	280 (62%)	119 (64%)	112 (61%)
Radiotherapy of the brain	25 (6%)	27 (6%)	14 (8%)	11 (6%)
Time from last surgery until diagnosis with unresectable locally advanced or metastatic disease, months	24.5 (15.9–38.9)	24.8 (17.0–44.6)	21.5 (15.0–36.2)	22.1 (14.1–39.3)

Adapted from Schmid and colleagues,²¹ by permission of Massachusetts Medical Society. Data are median (IQR), n (%), or n/N (%). Summary statistics are based on the full population indicated in the header, unless otherwise indicated. If the baseline characteristic was not available for all patients, the total number of patients evaluable for this characteristic is shown. *Two patients (one in each group) had a performance status of 2 before the start of treatment; they had a lower performance status at randomisation, hence why they were included in the study. †As recorded in the case report form.

Table 1: Baseline characteristics

survival between treatment groups was tested with an allocated type I error at an α level of at least 0.04 for overall survival and hierarchical testing for overall survival, first in the intention-to-treat population and then in the PD-L1 immune cell-positive subgroup (appendix p 7). Only if the difference in overall survival between the treatment groups in the intention-to-treat population was significant could overall survival in the

PD-L1 immune cell-positive subgroup be tested by re-using the type I error used for overall survival testing (≥ 0.04). The Kaplan-Meier method was used to analyse progression-free survival and overall survival, with the Brookmeyer-Crowley method used to estimate the 95% CI for each median duration. The proportional hazards assumption for overall survival was assessed in an exploratory manner. The same methods described for

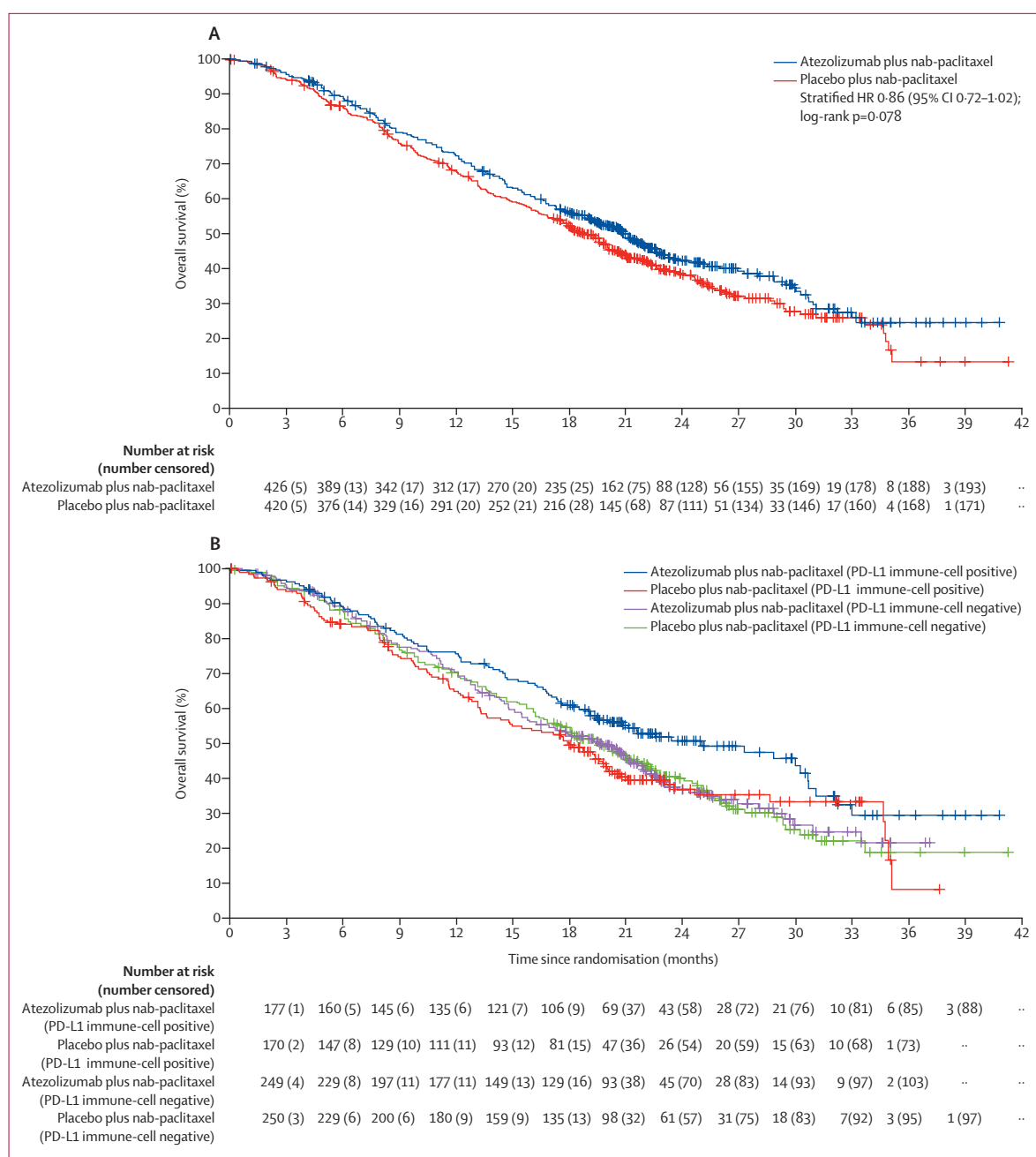


Figure 2: Second interim overall survival analysis

Kaplan-Meier analysis of overall survival in the (A) intention-to-treat and (B) PD-L1 immune cell-positive and PD-L1 immune cell-negative populations. HR is from stratified log-rank tests. HR=hazard ratio.

the analysis of progression-free survival were used for the analysis of duration of response except that the analysis was not stratified. Subgroup analyses of overall survival, prespecified in the statistical analysis plan, are provided. Interaction between PD-L1 status and overall survival benefit with atezolizumab plus nab-paclitaxel was tested using a stratified Cox-model including the covariate treatment group, PD-L1 status, and their interaction. Safety was evaluated in the safety population

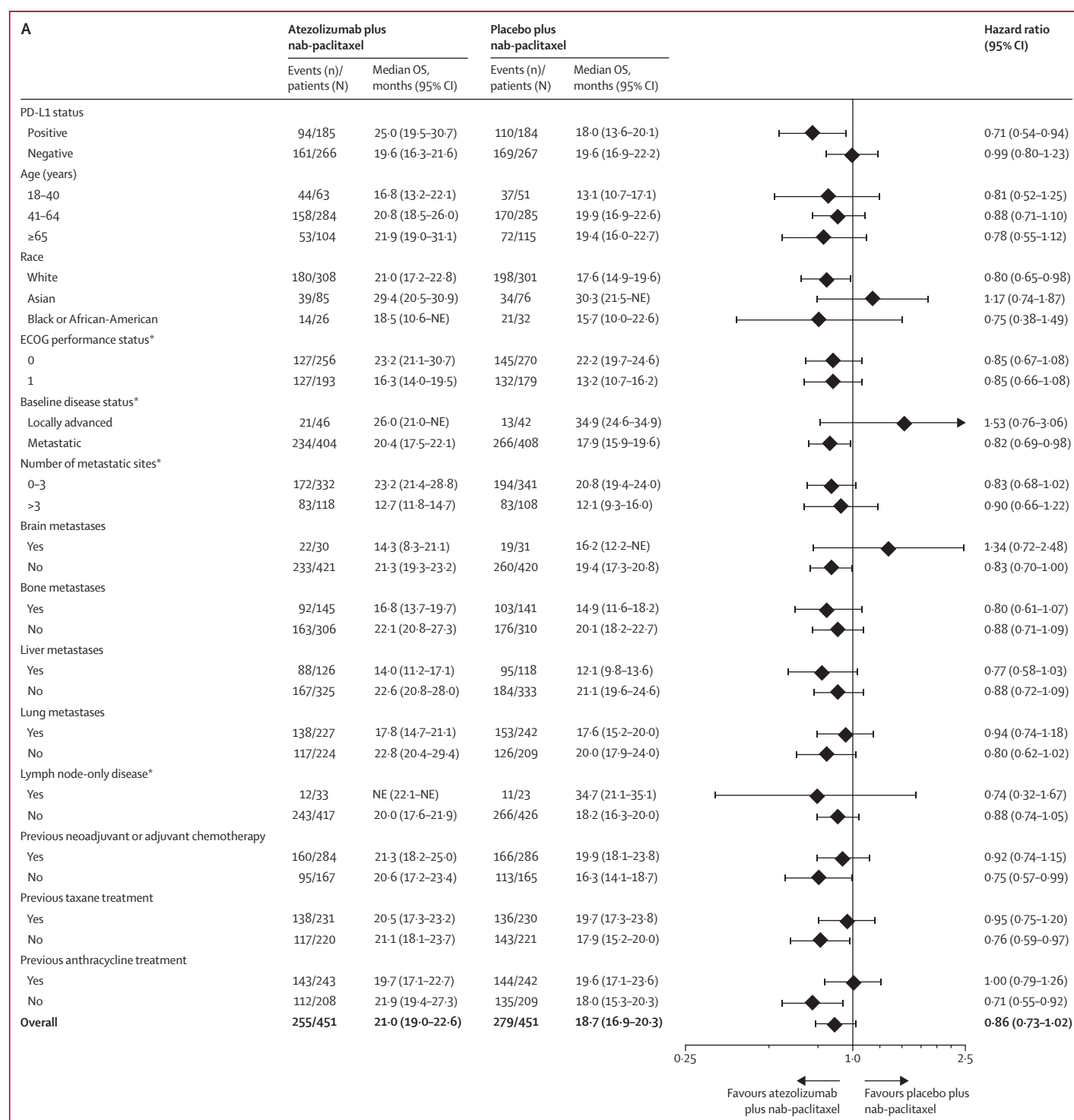
(ie, patients who received any amount of any study drug). SAS version 9.4 was used for all statistical analyses. This study is registered with ClinicalTrials.gov, NCT02425891.

Role of the funding source

The funders provided the study drug (atezolizumab) and the matched placebo, and collaborated with academic authors on the study design, data collection, data analysis,

data interpretation, and writing of the report. Celgene provided nab-paclitaxel, but the company had no role in study design, data collection, data analysis, or data interpretation, although they did review the manuscript. All authors contributed to drafting the manuscript,

provided final approval to publish, and agreed to be accountable for all aspects of the manuscript. VH, LM, SYC, VM, and AH had full access to the raw data and PS and HSR had final responsibility for the decision to submit for publication.



(Figure 3 continues on next page)

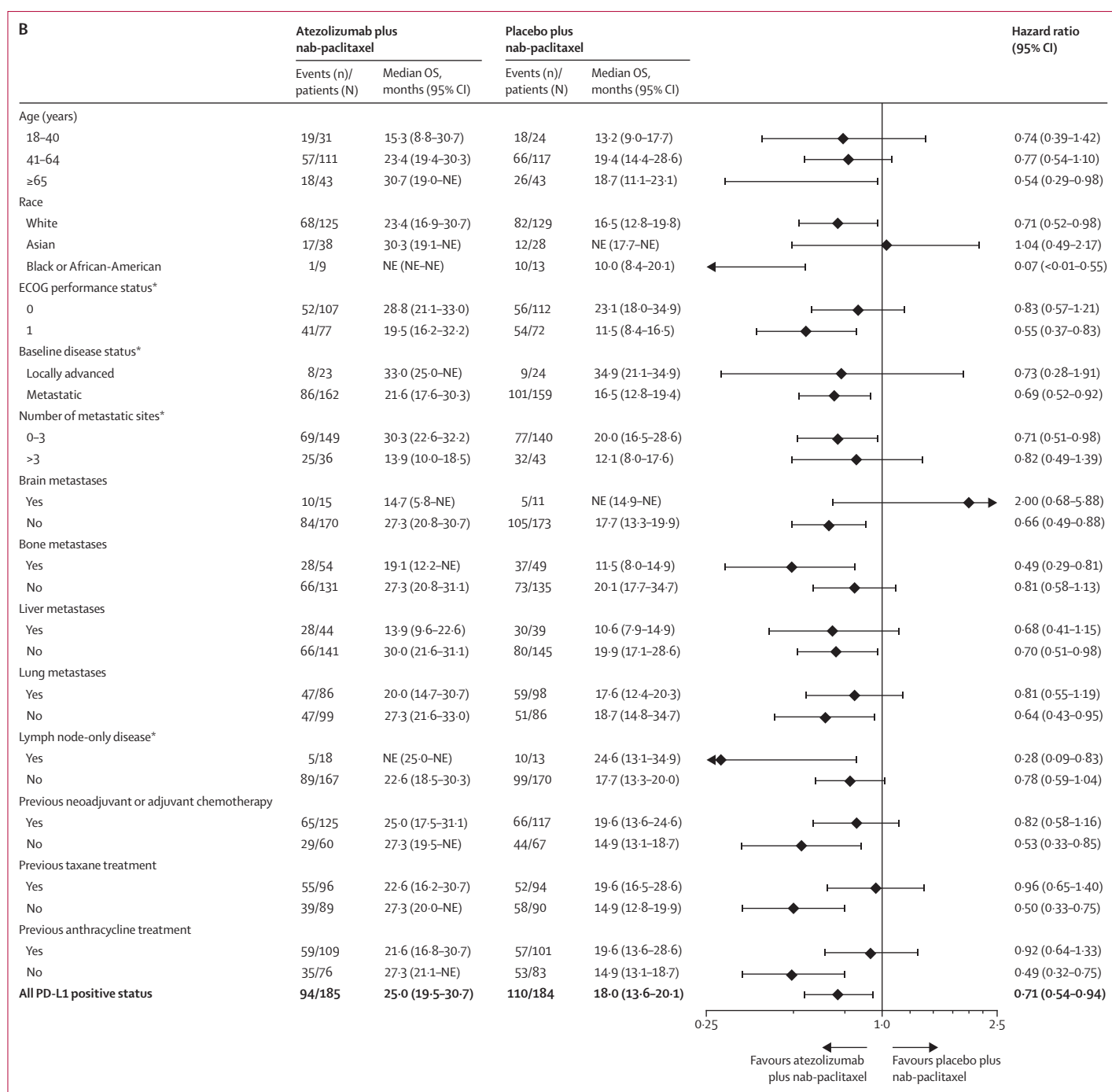


Figure 3: Overall survival in patient subgroups

Overall survival in subgroup populations in the (A) intention-to-treat and (B) PD-L1 immune cell-positive populations. ECOG=Eastern Cooperative Oncology Group. NE=not estimable. OS=overall survival. *Excludes patients with unknown or other values for indicated categories. The two enrolled patients with ECOG performance status 2 are also not plotted.

Results

Between June 23, 2015, and May 24, 2017, we enrolled 902 patients. The clinical cutoff for the primary progression-free survival and first interim overall survival analysis was on April 17, 2018. At the time of clinical data cutoff reported here (Jan 2, 2019), participants had been

followed up for a median of 18.5 months (IQR 9.6–22.8) in the atezolizumab group and 17.5 months (8.4–22.4) in the placebo group. 451 patients were assigned to atezolizumab plus nab-paclitaxel and 451 to placebo plus nab-paclitaxel and were included in the primary intention-to-treat efficacy analysis. Six (1%) patients in

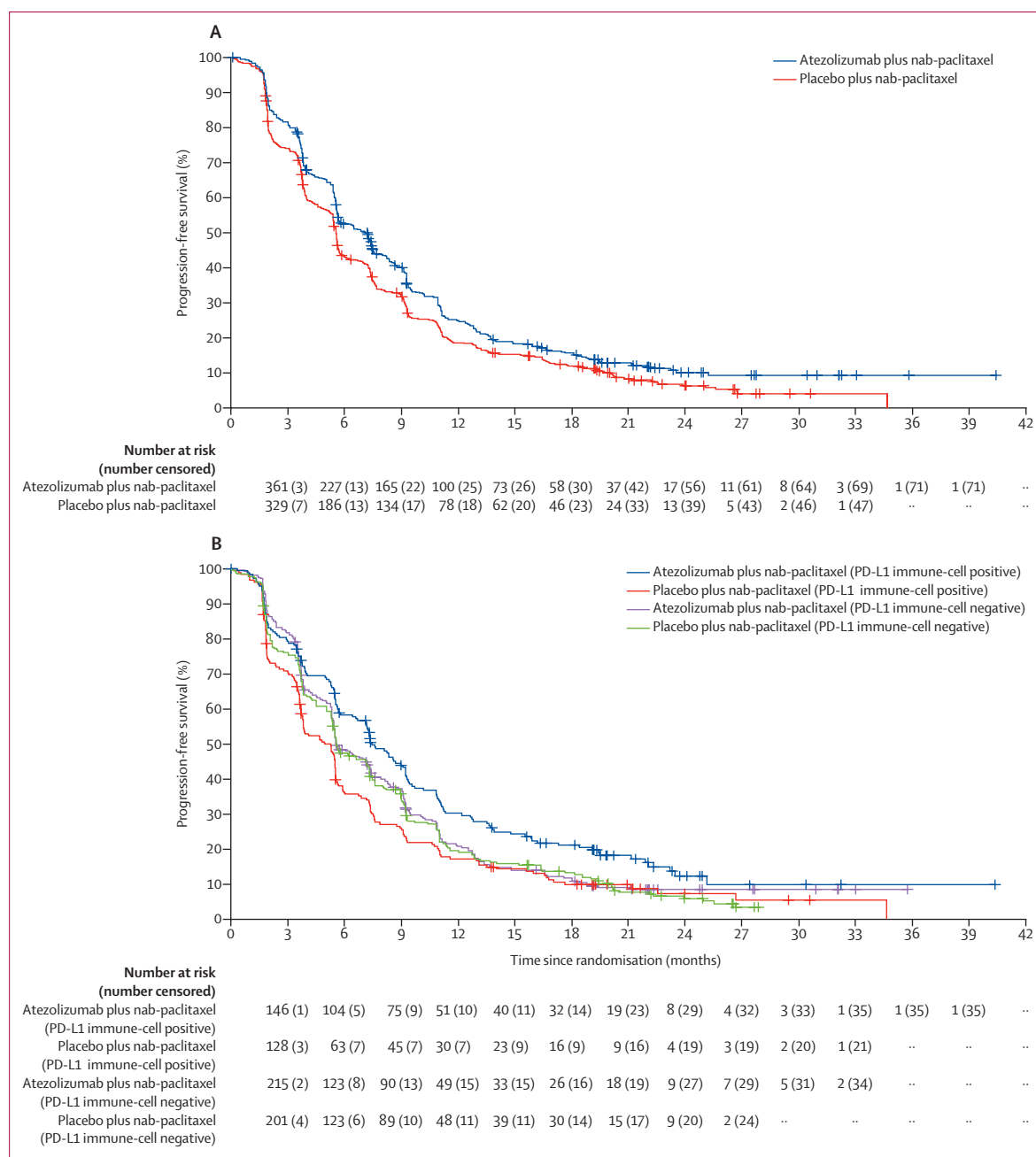


Figure 4: Progression-free survival

Kaplan-Meier analysis of progression-free survival in the (A) intention-to-treat and (B) PD-L1 immune cell-positive and PD-L1 immune cell-negative populations.

the atezolizumab group and six (1%) patients in the placebo group did not receive any treatment. Eight (2%) patients in the placebo group received one dose of atezolizumab and were therefore included in the safety-evaluable population for atezolizumab. Thus, the safety-evaluable population comprised 453 patients in the atezolizumab group and 437 in the placebo group (figure 1).

The patient baseline characteristics, as previously described,² were generally well balanced between treatment groups (table 1), and they were consistent between the intention-to-treat and PD-L1 immune cell-positive populations. Of the intention-to-treat population, 369 (41%) patients were PD-L1 immune cell positive (185 in the atezolizumab group and 184 in the placebo group).

	Atezolizumab plus nab-paclitaxel (n=453)	Placebo plus nab-paclitaxel (n=437)
Total number of reported adverse events of any cause	7741	6061
All-cause adverse events		
Any grade	450 (99%)	428 (98%)
Grade 3 or 4	224 (49%)	187 (43%)
Grade 5	6 (1%)*	3 (1%)†
Serious adverse events‡	105 (23%)	81 (19%)
Adverse events leading to discontinuation of any study drug	74 (16%)	35 (8%)
Adverse events leading to discontinuation of atezolizumab or placebo only	30 (7%)	5 (1%)
Adverse events leading to discontinuation of nab-paclitaxel only	74 (16%)	35 (8%)
Adverse events leading to dose reduction or interruption of nab-paclitaxel§	196 (43%)	173 (40%)
Treatment-related adverse events		
Any-grade adverse events	437 (97%)	410 (94%)
Grade 3 or 4	181 (40%)	133 (30%)
Grade 5	2 (<1%)¶	1 (<1%)
Serious adverse events	57 (13%)	32 (7%)
Adverse events of special interest**		
Any-grade adverse events of special interest	262 (58%)	183 (42%)
Grade 3 or 4	36 (8%)	19 (4%)
Grade 5	1 (<1%)††	1 (<1%)‡‡
Adverse events leading to atezolizumab or placebo discontinuation	9 (2%)	2 (<1%)
Treated with systemic corticosteroids within 30 days of onset of the adverse event of special interest	62 (14%)	28 (6%)

Data are n (%). *Aspiration (n=1), pneumonia (n=1), autoimmune hepatitis (n=1), (unexplained) death (n=1), pulmonary embolism (n=1), and septic shock (n=1). †Hepatic failure (n=1), acute myocardial infarction (n=1), and (unexplained) death (n=1). ‡For serious adverse events with prevalence of 1% or more see appendix (p 9). §Dose reductions of atezolizumab or placebo were not permitted as per protocol. ¶Atezolizumab-related autoimmune hepatitis (n=1) and septic shock related to nab-paclitaxel (n=1). ||Hepatic failure related to either drug (n=1). **As predefined by the study funder. ††Autoimmune hepatitis (n=1). ‡‡Hepatic failure (n=1).

Table 2: Safety summary

At the time of clinical data cutoff for our analysis (Jan 2, 2019), 255 (57%) of 451 patients in the atezolizumab group and 279 (62%) of 451 patients in the placebo group had died. In the intention-to-treat population, median overall survival was 21·0 months (95% CI 19·0–22·6) with atezolizumab and 18·7 months (16·9–20·3) with placebo (figure 2A). The stratified HR for overall survival was 0·86 (95% CI 0·72–1·02, $p=0·078$), with a p value of 0·070 for the interaction between PD-L1 status and overall survival benefit with atezolizumab. In a post-hoc analysis (for which timepoints were planned before unblinding of the study), 24-month overall survival was 42·4% (95% CI 37·3–47·4) in the atezolizumab group and 38·7% (33·7–43·6) in the placebo group. Visual inspection of the Kaplan-Meier curves did not indicate any non-proportionality in the hazard ratio.

Because PD-L1 immune cell-positive status predicts clinical benefit with atezolizumab plus nab-paclitaxel (as reflected in health authority approvals for patients with PD-L1-positive tumours), we did exploratory

analyses in this population, even though overall survival in the intention-to-treat population was not significantly different between the treatment groups. In patients with PD-L1-positive tumours, median overall survival was 25·0 months (95% CI 19·6–30·7) with atezolizumab and 18·0 months (13·6–20·1) with placebo (figure 2B). The stratified HR for overall survival was 0·71 (95% CI 0·54–0·94). 24-month overall survival was 50·7% (95% CI 42·9–58·5) in the atezolizumab group versus 36·9% (29·0–44·9) in the placebo group. Median overall survival in the PD-L1 immune cell-negative population was 19·7 months (95% CI 16·3–21·6) with atezolizumab and 19·6 months (16·9–22·2) with placebo (stratified HR 0·97; 95% CI 0·78–1·20). Subgroup analyses of overall survival in the intention-to-treat and PD-L1 immune cell-positive populations are shown in figure 3.

The final progression-free survival was already done at the first interim analysis of overall survival on April 17, 2018. However, because more progression-free survival event information had been collected at the time of the second interim overall survival analysis, we present a post-hoc update to better characterise the long-term progression-free survival results, especially in the PD-L1 immune cell-positive patient population. At the time of clinical data cutoff for this second analysis (Jan 2, 2019), 379 (84%) of 451 patients in the atezolizumab group and 404 (90%) of 451 patients in the placebo group had disease progression or had died. In the intention-to-treat population, median progression-free survival was 7·2 months (95% CI 5·6–7·4) with atezolizumab and 5·5 months (5·3–5·6) with placebo (stratified HR 0·80 [95% CI 0·69–0·92], $p=0·0021$; figure 4A). 24-month progression-free survival was 10% (95% CI 7–13) in the nab-paclitaxel group and 6% (4–9) in the placebo group. Sites of new lesions per investigator assessment are shown in the appendix p 8.

In the PD-L1 immune cell-positive population, the updated median progression-free survival was 7·5 months (95% CI 6·7–9·2) with atezolizumab and 5·3 months (3·8–5·6) with placebo (stratified HR 0·63 [95% CI 0·50–0·80], $p<0·0001$; figure 4B). 12-month progression-free survival was 30·3% (95% CI 23·5–37·2) in the atezolizumab group and 17·3% (11·7–22·9) in the placebo group, and 24-month progression-free survival was 12·4% (95% CI 6·5–18·3) versus 7·4% (2·8–12·0), respectively. In the PD-L1 immune cell-negative patients, median progression-free survival was 5·6 months (95% CI 5·5–7·3) in the atezolizumab group versus 5·6 months (5·4–7·3) in the placebo group (HR 0·93 [95% CI 0·77–1·11]). Final analyses of the secondary efficacy endpoints objective responses and duration of response have previously been reported.²

The safety update with an additional 4·5 months of follow-up since the primary analysis (data cutoff Sept 3, 2018) are summarised in tables 2 and 3. The 890 safety-evaluable patients were followed up for safety

	Atezolizumab plus nab-paclitaxel group (n=453)				Placebo plus nab-paclitaxel group (n=437)			
	Grade 1–2	Grade 3	Grade 4	Grade 5	Grade 1–2	Grade 3	Grade 4	Grade 5
Any adverse event	223 (49%)	197 (44%)	24 (5%)	6 (1%)	240 (55%)	157 (36%)	28 (6%)	3 (1%)
Alopecia	256 (57%)	2 (<1%)	0	0	249 (57%)	1 (<1%)	0	0
Nausea	204 (45%)	5 (1%)	0	0	160 (37%)	8 (2%)	0	0
Fatigue	195 (43%)	17 (4%)	0	0	180 (41%)	15 (3%)	0	0
Diarrhoea	139 (31%)	8 (2%)	0	0	143 (33%)	9 (2%)	0	0
Cough	114 (25%)	0	0	0	85 (20%)	0	0	0
Constipation	113 (25%)	3 (1%)	0	0	106 (24%)	1 (<1%)	0	0
Anaemia	112 (25%)	14 (3%)	0	0	105 (24%)	13 (3%)	0	0
Headache	106 (23%)	2 (<1%)	0	0	91 (21%)	4 (1%)	0	0
Decreased appetite	87 (19%)	3 (1%)	0	0	77 (18%)	3 (1%)	0	0
Vomiting	85 (19%)	4 (1%)	0	0	70 (16%)	5 (1%)	0	0
Pyrexia	85 (19%)	3 (1%)	0	0	48 (11%)	0	0	0
Arthralgia	81 (18%)	1 (<1%)	0	0	70 (16%)	1 (<1%)	0	0
Rash	76 (17%)	2 (<1%)	0	0	71 (<16%)	2 (1%)	0	0
Neuropathy peripheral	74 (16%)	25 (6%)	0	0	86 (20%)	12 (3%)	0	0
Dyspnoea	69 (15%)	3 (1%)	0	0	62 (14%)	3 (1%)	0	0
Peripheral oedema	68 (15%)	1 (<1%)	0	0	62 (14%)	6 (1%)	0	0
Back pain	66 (15%)	6 (1%)	0	0	57 (13%)	2 (1%)	0	0
Myalgia	65 (14%)	2 (<1%)	0	0	64 (15%)	3 (1%)	0	0
Dysgeusia	65 (14%)	0	0	0	59 (14%)	0	0	0
Dizziness	64 (14%)	0	0	0	46 (11%)	0	0	0
Peripheral sensory neuropathy	63 (14%)	9 (2%)	0	0	44 (10%)	8 (2%)	0	0
Pruritus	63 (14%)	0	0	0	45 (10%)	0	0	0
Hypothyroidism	63 (14%)	0	0	0	16 (4%)	0	0	0
Neutropenia	59 (13%)	28 (6%)	10 (2%)	0	32 (7%)	23 (5%)	13 (3%)	0
Asthenia	54 (12%)	3 (1%)	0	0	46 (11%)	4 (1%)	0	0
Urinary tract infection	52 (12%)	4 (1%)	0	0	43 (10%)	3 (1%)	0	0
Insomnia	52 (12%)	0	0	0	50 (11%)	3 (1%)	0	0
Pain in extremity	50 (11%)	2 (0.4)	0	0	42 (10%)	1 (<1%)	0	0
Nasopharyngitis	48 (11%)	0	0	0	37 (9%)	0	0	0
Abdominal pain	45 (10%)	2 (<1%)	0	0	52 (12%)	1 (<1%)	0	0
Aspartate aminotransferase increased	36 (8%)	8 (2%)	1 (<1%)	0	34 (8%)	8 (2%)	1 (<1%)	0
Neutrophil count decreased	35 (8%)	17 (4%)	5 (1%)	0	33 (8%)	13 (3%)	3 (1%)	0
Hypertension	18 (4%)	4 (1%)	0	0	15 (3%)	10 (2%)	1 (<1%)	0
Hypokalaemia	17 (4%)	8 (2%)	3 (1%)	0	6 (1%)	4 (1%)	0	0
Pneumonia	12 (3%)	10 (2%)	0	1 (<1%)	4 (1%)	3 (1%)	0	0

Data are n (%). All grade 1–2 adverse events that occurred in at least 10% of patients in either group, and grade 3–5 events that occurred in at least 2% of patients in either group, are shown. A full list of all adverse events in the safety population is in appendix pp 10–14.

Table 3: Adverse events in the safety population

for a median duration of 15.6 months (IQR 8.8–20.0). Treatment duration and exposure at the time of the safety update are shown in the appendix (p 9).

In the atezolizumab, 224 (49%) of 453 patients had grade 3 or 4 adverse events compared with 187 (43%) of 437 patients in the placebo plus nab-paclitaxel group. The most common grade 3–4 adverse events were neutropenia (38 [8%] of 453 patients in the atezolizumab group vs 36 [8%] of 437 patients in the placebo group), peripheral neuropathy (25 [6%] vs 12 [3%]), decreased neutrophil count (22 [5%] vs 16 [4%]), and fatigue (17 [4%] vs 15 [3%]). As was the case in the primary analysis, all grade 3–4

adverse events were balanced across treatment groups, with the exception of peripheral neuropathy (grade 3 only; 25 [6%] of 453 patients in the atezolizumab group vs 12 [3%] of 437 patients in the placebo group). The grade 3 peripheral neuropathy was deemed to be taxane-related, which is known to be cumulative. No cases of Guillain-Barré syndrome were reported. 74 (16%) of 453 patients in the atezolizumab group had an adverse event leading to discontinuation of any study drug (vs 35 [8%] of 437 patients in the placebo group; appendix p 9). Adverse events leading to discontinuation of atezolizumab or placebo occurred in 30 (7%) patients in the atezolizumab

Atezolizumab plus nab-paclitaxel group (n=451)						Placebo plus nab-paclitaxel group (n=437)					
Adverse events of special interest	Resolved adverse events of special interest	First adverse event of special interest (any grade)	Median time to onset, months	Any grade	Adverse events of special interest leading to withdrawal of atezolizumab	Adverse events of special interest leading to withdrawal of placebo	Adverse events of special interest	Resolved adverse events of special interest	First adverse event of special interest (any grade)	Median time to onset, months	Any grade
Any grade	Grade 3 or 4	Any grade	Median duration, months	Any grade	Any grade	Any grade	Any grade	Grade 3 or 4	Any grade	Median duration, months	Any grade
Hypothyroidism	80 (18%)	0	3.8 (2.2-6.0)	38/80 (48%)	13.0 (2.2-NE)	0	20 (5%)	0	2.0 (1.8-6.5)	NE (1.9-NE)	0
Hyperthyroidism	21 (5%)	1 (<1%)	3.8 (1.9-5.5)	16/21 (76%)	1.9 (1.1-8.1)	0	6 (1%)	0	3.2 (0.5-1.5)	0.8 (0.5-1.5)	0
Pneumonitis	16 (4%)	2 (<1%)	4.7 (1.9-6.8)	12/16 (75%)	3.2 (1.5-5.8)	1 (<1%)	1 (<1%)	0	1.0 (1.0-1.0)	NE (NE-NE)	0
Rash	155 (34%)	4 (<1%)	1.3 (0.4-4.0)	126/155 (81%)	1.0 (0.3-6.3)	1 (<1%)	115 (26%)	2 (<1%)	1.0 (0.4-2.8)	1.0 (0.4-7.2)	0
Hepatitis (diagnosis)*	11 (2%)	7 (2%)	4.6 (2.9-8.7)	6/11 (55%)	2.0 (1.0-NE)	2 (<1%)	7 (2%)	1 (<1%)	5.4 (2.3-7.8)	0.8 (0.7-1.0)	1 (<1%)
Colitis	5 (1%)	1 (<1%)	6.7 (4.9-8.1)	4/5 (80%)	0.8 (0.8-2.3)	1 (<1%)	3 (1%)	1 (<1%)	2.1 (1.4-6.4)	0.5 (0.2-1.4)	0
Adrenal insufficiency*	4 (<1%)	1 (<1%)	4.9 (4.7-8.9)	3/4 (75%)	5.5 (2.7-NE)	1 (<1%)	0	0	NA	NA	0

Data are n (%), n/N (%), or median (IQR). The table lists adverse events of special interest as medical concepts (grouped by Medical Dictionary for Regulatory Activities-preferred terms) occurring in more than ten patients in the atezolizumab plus nab-paclitaxel group, excluding hepatitis (laboratory abnormalities), and additionally including colitis and adrenal insufficiency. NE=not estimable. NA=not applicable. *Funder-defined group of terms that represent events suggestive of a hepatitis diagnosis (as opposed to events associated with liver function test abnormalities only).

Table 4: Clinically relevant adverse events of special interest

group and in five (1%) patients in the placebo group. Consistent with the primary analysis, the leading cause for treatment discontinuation of atezolizumab plus nab-paclitaxel due to toxicity was peripheral neuropathy (in 20 [4%] patients).

Serious adverse events were reported in 105 (23%) of 453 patients in the atezolizumab group compared with 81 (19%) of 437 patients in the placebo group (appendix p 9).

262 (58%) of 453 patients in the atezolizumab group and 183 (42%) of 437 in the placebo group had an adverse event of special interest (table 2, table 4). One death due to an adverse event of special interest occurred in each treatment group (hepatitis in both cases). Consistent with the primary analysis, the adverse events of special interest that differed substantially between the atezolizumab group and the placebo group were any-grade rash, hypothyroidism, hyperthyroidism, pneumonitis, and adrenal insufficiency (table 4). Adverse events leading to death occurred in six (1%) patients in the atezolizumab group and three (1%) patients in the placebo group. Treatment-related deaths occurred in two (<1%) patients in the atezolizumab group (one due to autoimmune hepatitis related to atezolizumab and one due to septic shock related to nab-paclitaxel only) and one (<1%) patient in the placebo group (hepatic failure). No new treatment-related deaths have been reported since the primary clinical data cutoff date (April 17, 2018).

274 (61%) of 451 patients from the atezolizumab group and 295 (65%) of 451 patients from the placebo group received at least one subsequent anticancer therapy (appendix pp 15–17). Antimetabolites (eg, capecitabine, gemcitabine, fluorouracil, and methotrexate) were the most common subsequent therapy, received by 191 (42%) of 451 patients from the atezolizumab group and 204 (45%) of 451 patients from the placebo group. Furthermore, 122 (27%) of 451 patients in each treatment group received at least one subsequent therapy with a platinum compound. 17 (4%) of 451 patients from the atezolizumab group and 26 (6%) of 451 patients from the placebo group received at least one subsequent therapy with a monoclonal antibody targeting PD-L1, PD-1, or CTLA-4.

Discussion

To our knowledge, IMpassion130 was the first phase 3 study reported that evaluated an antibody targeting the PD-L1–PD-1 signalling pathway in patients with unresectable, advanced or metastatic triple-negative breast cancer. In the primary analysis, IMpassion130 met its coprimary progression-free survival endpoint, showing a significant progression-free survival benefit with atezolizumab plus nab-paclitaxel compared with placebo plus nab-paclitaxel as a first-line treatment for patients with advanced or metastatic triple-negative breast cancer.² Since no treatment effect was seen in the PD-L1 immune cell-negative subgroup, the clinical

benefit of atezolizumab in the intention-to-treat population was driven by the improvement in progression-free survival in the PD-L1 immune cell-positive subgroup.²¹ The first prespecified interim overall survival analysis was aligned with the primary progression-free survival analysis and was immature with an information fraction of 59%. In this Article, we report the second prespecified interim overall survival analysis, a concurrent progression-free survival update, and a safety update for IMpassion130. These new data provide valuable information to health-care providers in the USA, EU, and other territories where atezolizumab plus nab-paclitaxel is an approved treatment regimen.

Consistent with the first interim overall survival analysis,² the results from this second interim overall survival analysis showed that although median overall survival in the intention-to-treat population was longer with the combination of atezolizumab plus nab-paclitaxel than with placebo plus nab-paclitaxel, the significance boundary was not crossed. Therefore, since median overall survival did not differ significantly between the treatment groups in this analysis, the statistical testing hierarchy did not allow for formal testing of overall survival in the PD-L1 immune cell-positive patient population. However, the magnitude of overall survival benefit with atezolizumab in PD-L1 immune-positive patients remained clinically meaningful, with an increase of 7 months in median overall survival with atezolizumab plus nab-paclitaxel compared with placebo plus nab-paclitaxel (HR 0·71 [95% CI 0·54–0·94]). With a median follow-up of 18·0 months and an information fraction of 80%, the reported overall survival results can now be considered mature. The event-driven final overall survival analysis will be reported in a subsequent publication.

A similar proportion of patients in both treatment groups (61% of those in the atezolizumab group and 65% of those in the placebo group) received post-protocol treatments. The remaining patients were either still receiving treatment or were not eligible for further treatment because of progression. In some cases, the second-line therapy was not recorded. The decision for or against post-protocol treatment was not recorded in our study. The most frequent post-protocol therapies, which might have affected overall survival in the study population, were antimetabolites. Very few patients received immune checkpoint inhibitors, and the proportion who received these drugs was similar between the two treatment groups.

The final primary progression-free survival analysis results of this trial were previously reported in 2018, with 358 (79%) of the 451 patients in the atezolizumab group and 378 (84%) of 451 in the placebo group having shown disease progression or died,² and the progression-free survival results were updated at this second interim overall survival analysis timepoint to further support the results of the primary analysis. The updated progression-free survival results in PD-L1 immune cell-positive patients support the previously reported progression-free

survival benefit with atezolizumab, with an improvement in median progression-free survival of 2·2 months with atezolizumab plus nab-paclitaxel compared with placebo plus nab-paclitaxel in this population.

The consistent results across several analyses of IMpassion130 have established the benefit of PD-L1 blockade with atezolizumab in combination with nab-paclitaxel as a first-line therapy in PD-L1 immune cell-positive unresectable, locally advanced or metastatic triple-negative breast cancer. Any overall survival benefit seemed to be limited to PD-L1 immune cell-positive patients, suggesting that the combination of atezolizumab plus nab-paclitaxel can effectively and specifically harness pre-existing anti-tumour immunity in patients with unresectable, locally advanced or metastatic PD-L1 immune cell-positive triple-negative breast cancer. Consistent with the benefit pattern observed in other cancer immunotherapy studies,^{26,27} the absolute benefit in overall survival was greater than the absolute benefit for progression-free survival in IMpassion130.

No new safety signals were observed, and toxicity data remained consistent with those from the previously published progression-free survival analysis.² No cumulative toxicities or new or late-onset safety signals were seen with longer follow-up. Potentially immune-mediated adverse events of special interest, a toxicity concern with checkpoint inhibitors as a class,²⁸ were manageable, but represent a new class of toxicity that is important to identify and manage appropriately. The median times to onset of adverse events of special interest were consistent with those observed in atezolizumab monotherapy trials.²³

The limitations of this study include the ineligibility of patients who had received previous radiotherapy or chemotherapy (including taxanes) in the curative setting less than 12 months before randomisation for this study. Findings from this study should therefore not be extrapolated to the excluded patient population with rapid disease progression. Furthermore, the statistical testing hierarchy did not allow us to do formal statistical testing of overall survival results in the PD-L1 immune cell-positive population because the requirement for a significant overall survival improvement in the intention-to-treat population was not met.

In conclusion, our analysis indicates that there might be a clinically meaningful benefit of using atezolizumab in combination with nab-paclitaxel as a first-line treatment for patients with PD-L1 immune cell-positive unresectable, locally advanced or metastatic triple-negative breast cancer. The atezolizumab plus nab-paclitaxel combination is, to the best of our knowledge, the first therapy to cross the 2-year landmark overall survival benefit in PD-L1 immune cell-positive metastatic triple-negative breast cancer. Routine testing for PD-L1 immune-cell expression in unresectable, locally advanced or metastatic triple-negative breast cancer with the approved companion diagnostic (VENTANA PD-L1 SP142 assay) should be used

to identify patients who might benefit from treatment with atezolizumab plus nab-paclitaxel.

Contributors

PS, HSR, SA, AS, HI, VD, LM, SYC, AH, SL, and LAE designed the study. PS, AS, VH, LM, SYC, AH, and SL developed the study methods. PS, HSR, SA, AS, CHB, HI, VD, VH, SYC, AH, SL, and LAE collected data. VH, LM, SYC, and AH had access to the raw data. PS, HSR, SA, AS, CHB, HI, VD, VH, LM, SYC, VM, AH, EPW, SL, and LAE analysed and interpreted the data. All authors contributed to manuscript writing, approved the final version, and are accountable for all aspects of the report. All authors contributed to drafting the manuscript, provided final approval to publish, and agree to be accountable for all aspects of the manuscript.

Declaration of interests

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Data sharing

Qualified researchers can request access to individual patient-level data through the clinical study data request platform. Details on Roche's criteria for eligible studies are available online. Those interested in accessing study data should view Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical study documents.

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